

Why the SxL?

The Spatial Experience Lab was chosen as a space where the specific problem PointoBot Investigates occurs for real users.

UNFAMILIARITY IS BUILT IN

The lab reconfigures constantly. Even frequent users face spatial confusion on every new setup.

SPATIAL KNOWLEDGE IS TASK-CRITICAL

People need to know specific locations of specific things to do their work - cables, boundaries, coordinate origins.

LANGUAGE CONSISTENTLY FAILS

There is no adequate way to describe where things are. Pointing at a diagram is not the same as pointing in the room.

INDEPENDENT USE MODEL

No constant technical support. Users must resolve spatial problems themselves - that is exactly the gap PointoBot fills.

USER SURVEY EVIDENCE

People often leave their setups there, and I'm not always sure whether it's okay to move them around - so I've been leaving tape marks on the floor if I have to move them.

EQUIPMENT LOCATION + OWNERSHIP

WANDER TO FIND

TRAIL & ERROR

It's hard to understand where the corners of the projectors are in Unity and align the camera accordingly.

CO-ORDINATE
SYSTEM ALIGNMENT

ASK A PEER

WAIT FOR RA

Finding where the tracking in Unity lines up with the room so that the projector and the markers fine up exactly - in both the digital and virtual space.

PROJECTOR BOUNDARY
ALIGNMENT

TRAIL & ERROR